



Impact of Internet Usage on Unemployment and Employment Structure

Course: DARP

Group: 7

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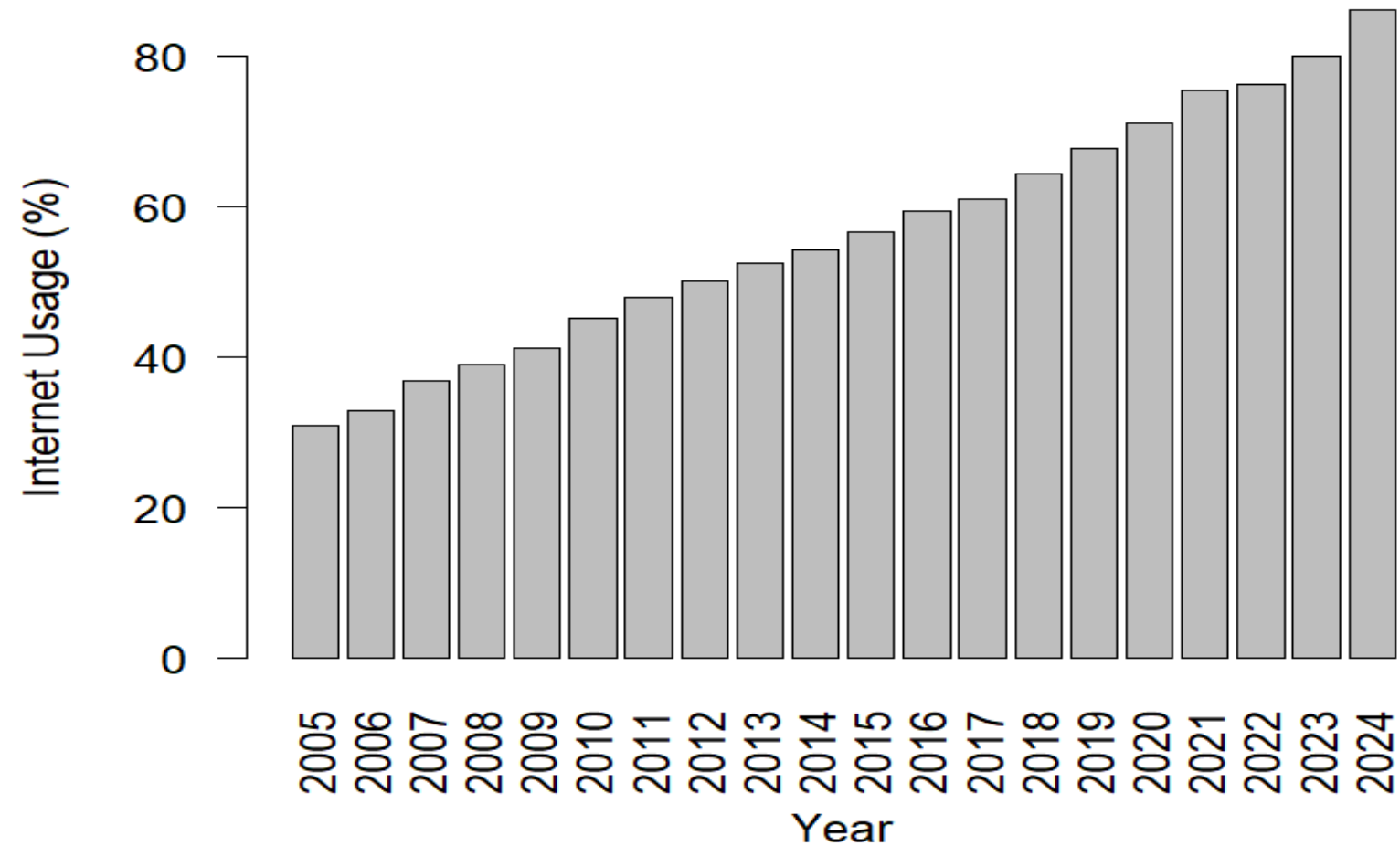
- Internet is a **key part of the modern economy**
- Supports **communication, education, business activities, and job search**
- Internet usage has **increased globally over time**
- Labour markets changed due to **technology and COVID-19**
- Study examines impact on **unemployment and employment structure**

- Study relationship between **internet usage and unemployment**
- Analyse impact on **employment structure**
- Examine the role of **GDP and other variables**
- Analyse impact during **COVID-19 period**
- Evaluate effects across **employment types**

- **Source:** World Bank (WDI)
- **Coverage:**
 - Countries: **18**
 - Period: **2005–2024**
 - Observations: **~360**
- **Variables:**
 - Internet usage (% population)
 - GDP growth & unemployment rate
 - Youth unemployment
 - Employment types:
 - NEET
 - Wage & salaried employment
 - Self–employment
 - Vulnerable employment
 - Services employment

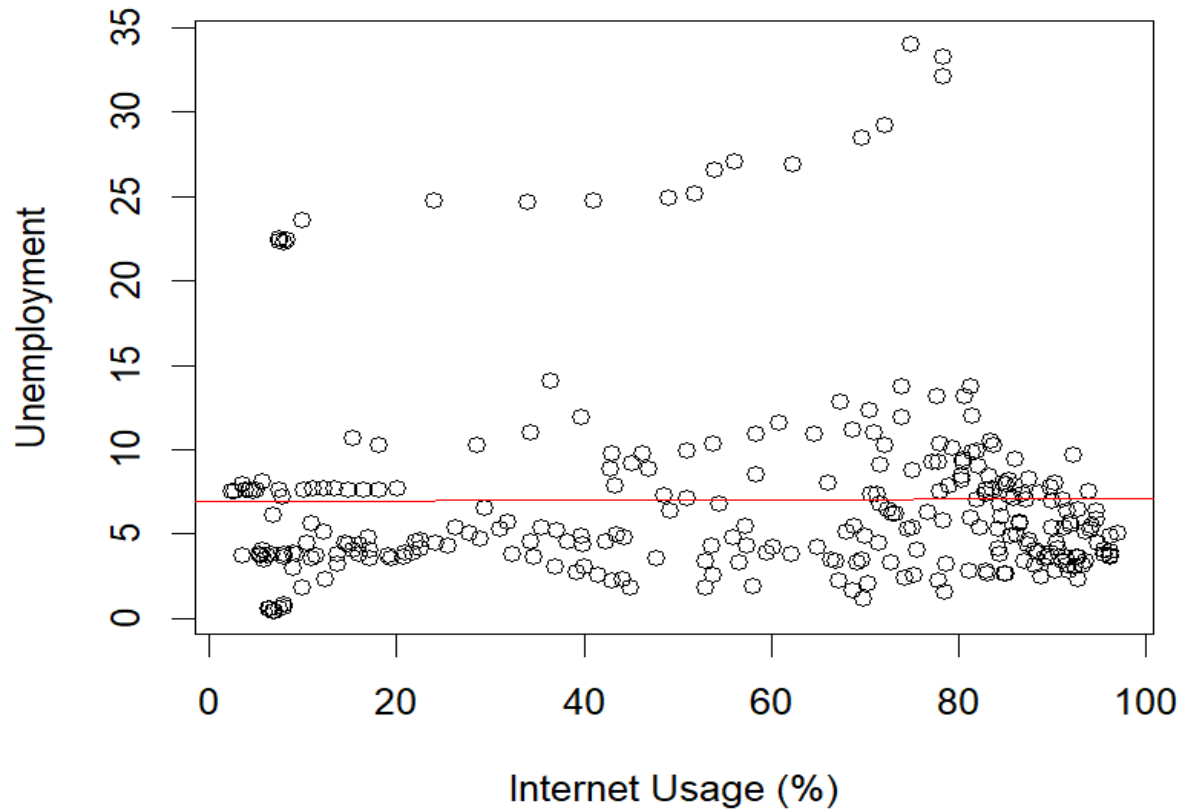
- Dataset was cleaned using **dplyr** and **tidyr**
- Rows and columns were rearranged where required
- Unnecessary variables such as country codes were removed
- Blank cells and symbols were replaced with **NA**

Internet Usage Over Time



YEAR	AVERAGE INTERNET USAGE(%)
2005	~31
2020	~71
2024	~86

Internet Usage and Unemployment



Key Result:

- $R^2 = 0.0003$
- Very weak relationship

Insight:

- Internet usage alone does not explain unemployment

“This suggests that other economic factors are important in explaining unemployment.”

Multiple Linear Regression

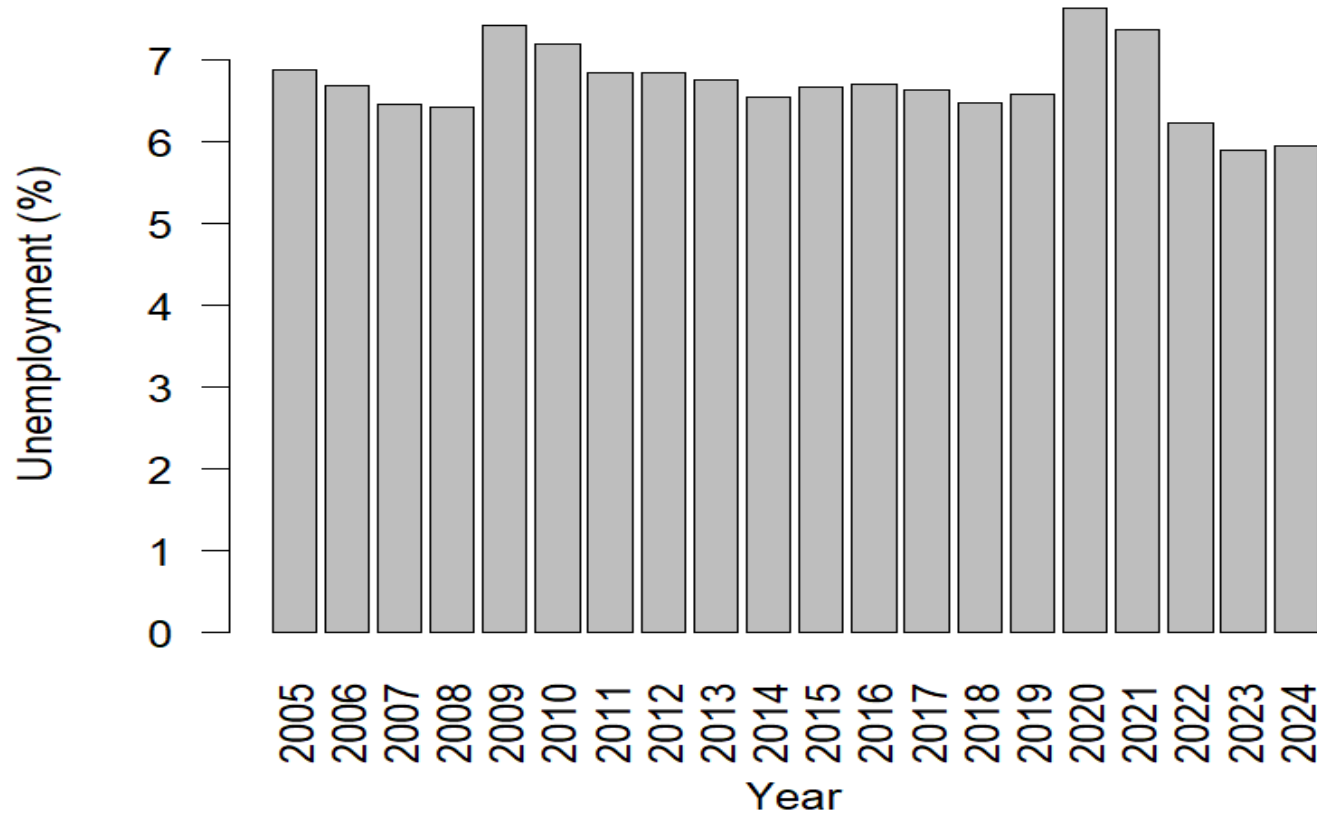
Variable	Coefficient	Effect
Internet Usage	-0.0996	Reduces unemployment
GDP	-0.3259	Strong reduction
Education	0.1352	Increases unemployment
Urban	0.0005	No significant effect
Intercept	1.3521	Base level

Model Fit: $R^2 = 0.145$

After adding GDP, education, and urbanization, internet usage shows a stronger negative relationship with unemployment than in the simple regression model.

Impact of COVID-19 on Unemployment

Trend of Unemployment Over Time



Period	Observation
2020–2021	Sharp increase
Later Years	Gradual recovery

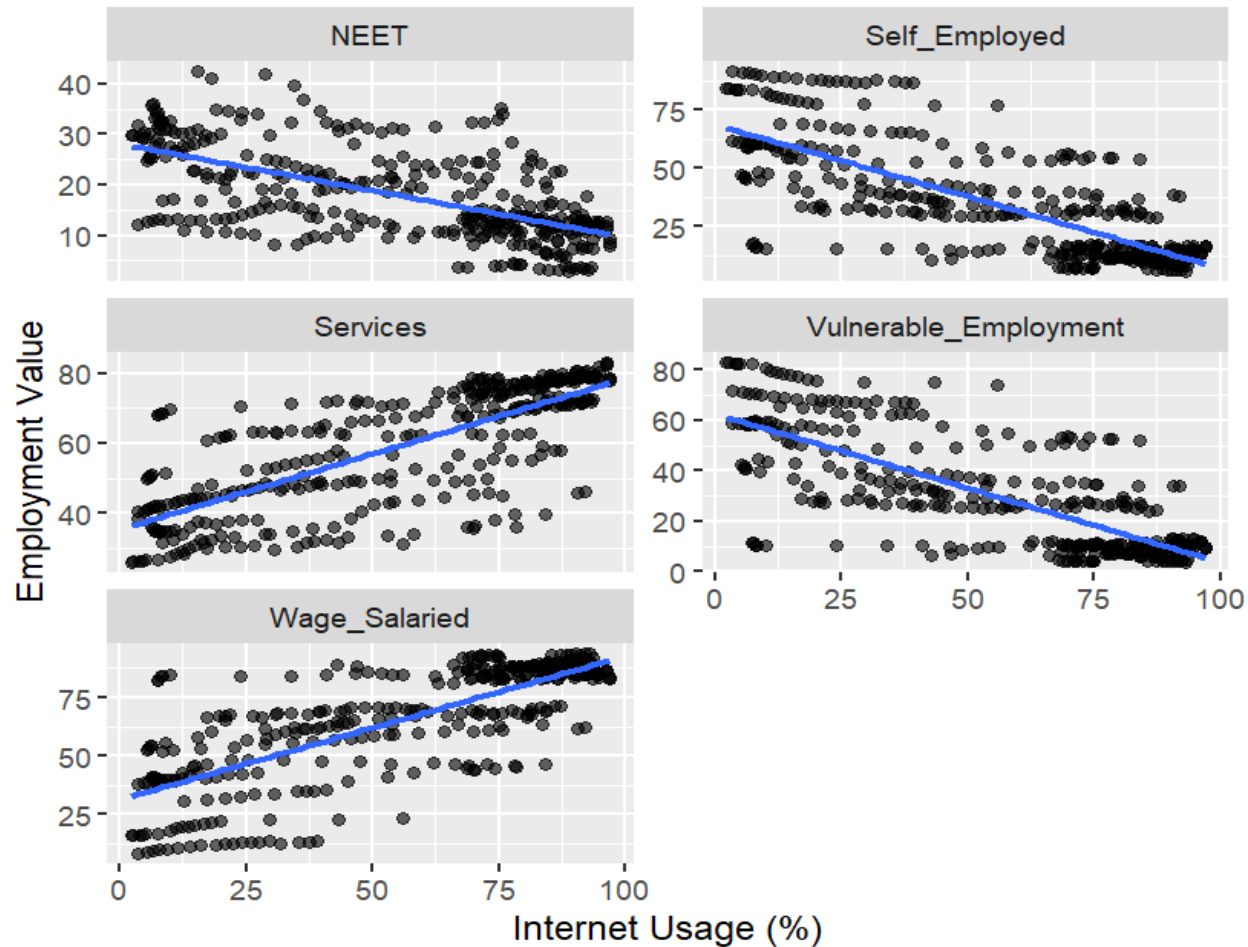
“Unemployment increased sharply during COVID-19 due to economic disruption, but declined in later years as recovery began.”

Impact of Internet Usage on Youth Outcomes

Indicator	Expected Impact of Better Internet Access
Youth Unemployment	Lower through job search opportunities
NEET	Lower through training and education access

“Internet usage helps improve youth outcomes by reducing unemployment and increasing access to education and job opportunities.”

Internet Usage and Employment Structure



“Greater internet access reduces labor market exclusion (NEET) and drives a transition toward formal, service-oriented, and less vulnerable employment.”

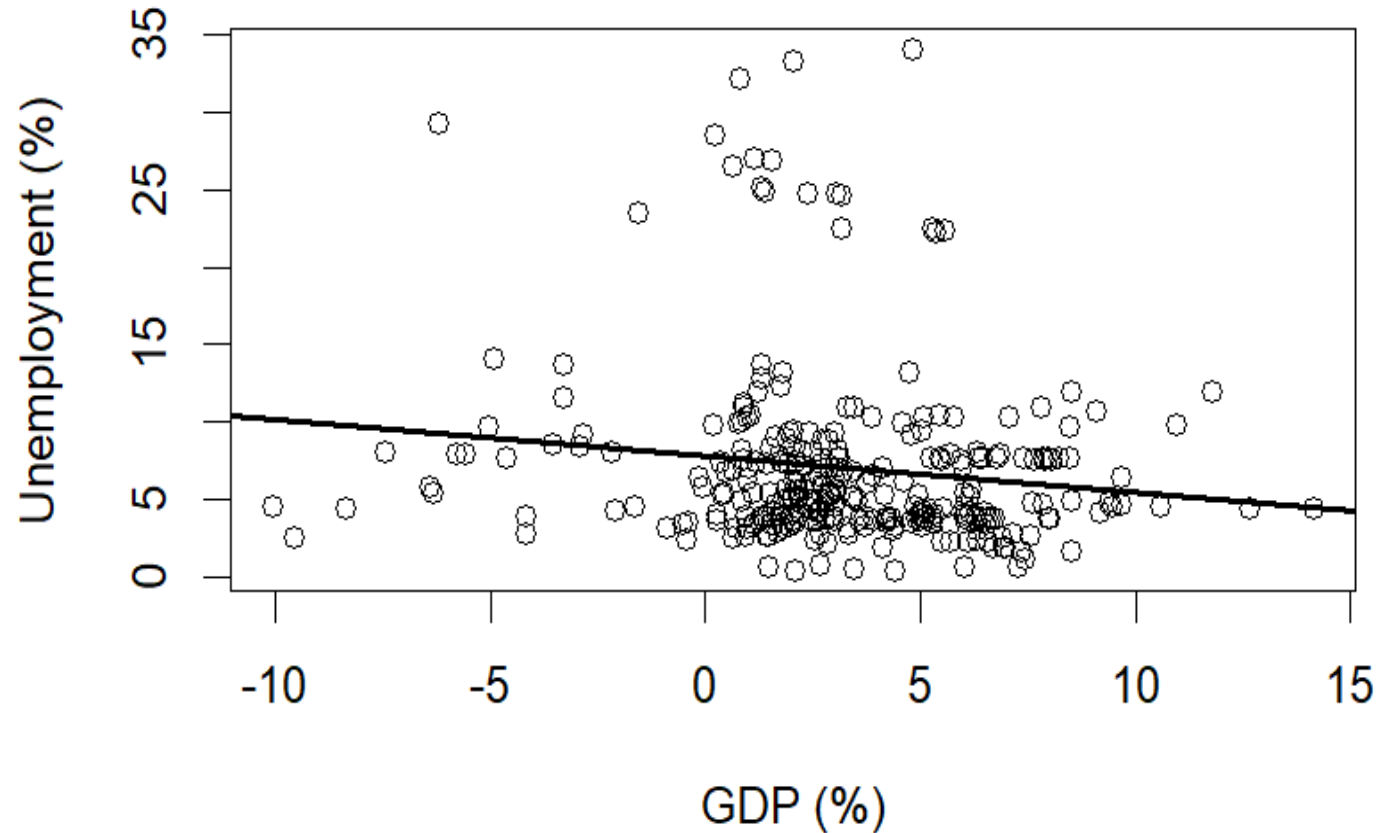
Impact on Employment Structure

Employment Type	Coefficient	Effect
Neet	-0.1851	↓
Self-Employment	-0.6146	↓
Services Employment	0.4325	↑
Vulnerable Employment	-0.5908	↓
Wage & Salaried Employment	0.6146	↑

Model Strength: $R^2 \sim 0.57-0.58$

”The regression results confirm a strong relationship, indicating that internet usage significantly improves employment structure.”

Impact of GDP Growth on Unemployment



Variable	Coefficient	Effect
GDP Growth (annual %)	-0.238	Reduces unemployment

”Higher GDP growth is associated with lower unemployment, indicating that economic expansion plays a key role in job creation.”

CI: High vs Low level internet countries



- Countries divided into two groups:
 - High internet usage (above median)
 - Low internet usage (below median)

Results (95% Confidence Intervals):

- High-internet countries: **[5.97, 7.49]**
- Low-internet countries: **[5.84, 7.61]**

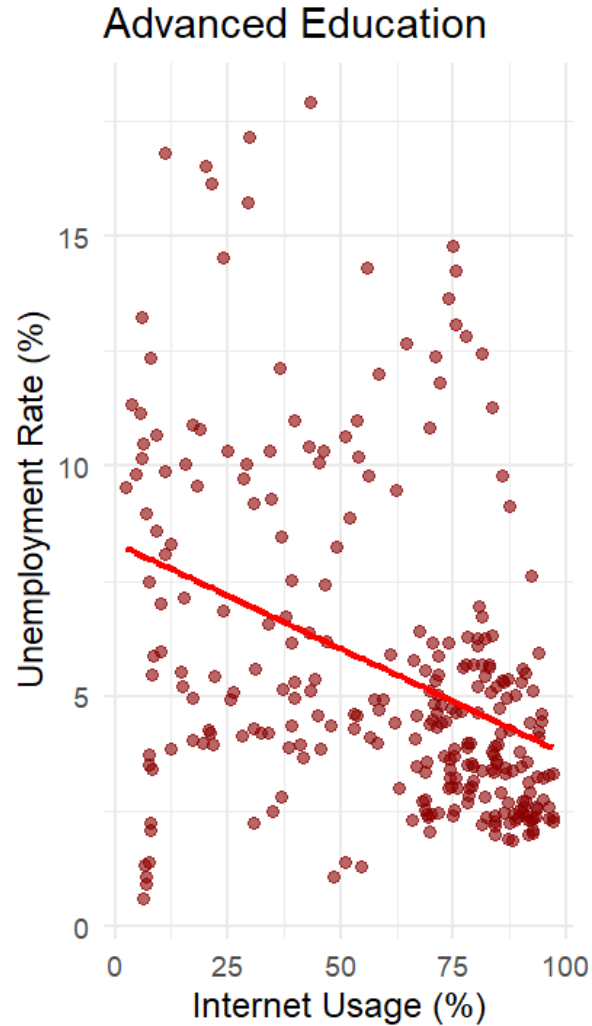
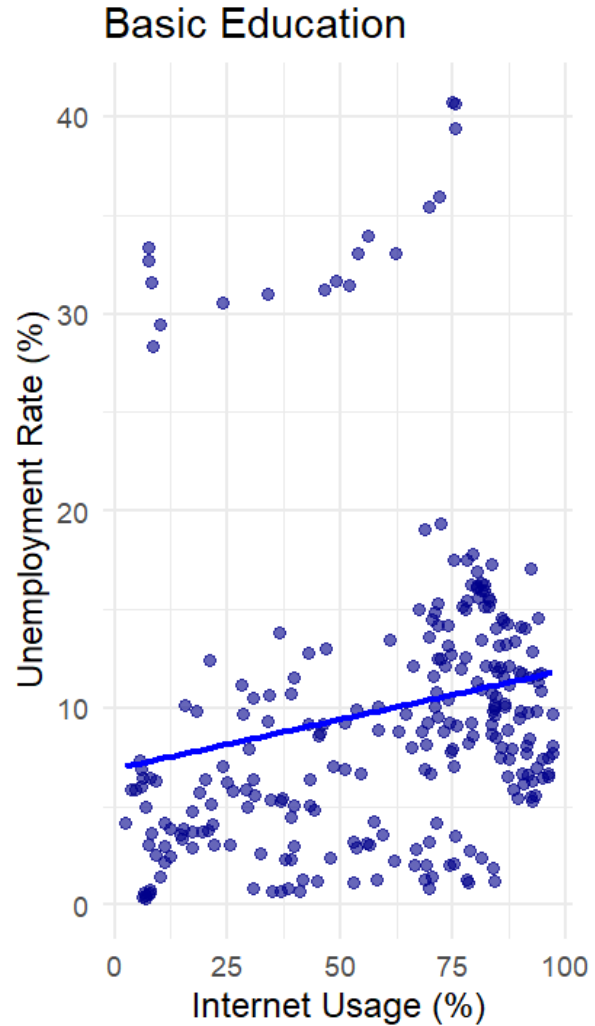
Key Insight:

- The intervals overlap significantly

Interpretation:

- Average unemployment levels are not clearly different
- Any observed difference may be due to random variation

Internet Usage & Education



Group	Coeff.	R ²	Result
Basic Edu.	0.0503	0.0339	Weak +ve
Advanced Edu.	-0.0459	0.1433	Negative

Internet access appears more beneficial for advanced-educated workers in labour markets.

Key Findings

- **Australia:** Highest internet usage → strong infrastructure and high income
- **Pakistan:** Lowest internet usage → infrastructure gaps and unequal access
- **South Africa:** High unemployment → structural issues and inequality
- **Vietnam:** Low unemployment → strong manufacturing and export-led growth
- **China:** Highest GDP growth → driven by industrialization and investment
- **Japan:** Lowest GDP growth → mature economy and aging population

“Better digital access and economic growth improve employment outcomes, while structural issues weaken them.”

- **Internet Usage:**
 - Rapid increase in internet penetration over time
 - Driven by affordable mobile data and smartphones
 - Improved access to online jobs, learning, and services
- **Unemployment:**
 - Unemployment pressure remained despite digital growth
 - Higher challenge among youth and educated workers
 - Informal sector employment still significant
- **GDP Growth:**
 - Strong GDP growth compared with many developed economies
 - Supported by services sector expansion
 - Large domestic market increased demand and production
 - Digital economy and technology sector contributed positively
 - COVID temporarily slowed growth, followed by recovery

- Internet usage alone has a weak relationship with unemployment.
- Including GDP and other variables improves explanation.
- Internet usage positively affects employment structure.
- Economic growth plays a key role in reducing unemployment.

“Overall, digitalization influences employment structure more than unemployment, while economic growth remains a key driver of job creation.”

Thank you for your attention.

We welcome any questions.